

Multi-Granularity Part Sampling Attention for Fine-Grained Visual Classification

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Abstract—Fine-grained visual classification aims to classify similar sub-categories with the challenges of large variations within the same sub-category and high visual similarities between different sub-categories. Recently, methods that extract semantic parts of the discriminative regions have attracted increasing attention. However, most existing methods extract the part features via rectangular bounding boxes by object detection module or attention mechanism, which makes it difficult to capture the rich shape information of objects. In this paper, we propose a novel Multi-Granularity Part Sampling Attention (MPSA) network for fine-grained visual classification. First, a novel multi-granularity part retrospect block is designed to extract the part information of different scales and enhance the high-level feature representation with discriminative part features of different granularities. Then, to extract part features of various shapes at each granularity, we propose part sampling attention, which can sample the implicit semantic parts on the feature maps comprehensively. The proposed part sampling attention not only considers the importance of sampled parts but also adopts the part dropout to reduce the overfitting issue. In addition, we propose a novel multi-granularity fusion method to highlight the foreground features and suppress the background noises with the assistance of the gradient class activation map. Experimental results demonstrate that the proposed MPSA achieves state-of-the-art performance on four commonly used fine-grained visual classification benchmarks. The source code is publicly available at <https://github.com/mobulan/MPSA>.

Index Terms—Fine-grained visual classification, multi-granularity, part sampling, attention mechanism.

Manuscript received 9 December 2023; revised 27 May 2024 and 8 July 2024; accepted 14 July 2024. Date of publication 16 August 2024; date of current version 22 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 6186206004, in part by the Natural Science Foundation of Anhui Province under Grant 2108085Y23, in part by the Natural Science Foundation for the Higher Education Institutions of Anhui Province under Grant KJ2021A0038, in part by the University Synergy Innovation Program of Anhui Province under Grant GXXT-2022-032, and in part by Anhui Provincial Key Research and Development Program under Grant 2022i01020014. The associate editor coordinating the review of this article and approving it for publication was Dr. Ananda Shankar Chowdhury. (Corresponding authors: Qin Xu; Bo Jiang.)

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Digital Object Identifier 10.1109/TIP.2024.3441813

I. INTRODUCTION

FINE-GRAINED visual classification (FGVC), which aims to distinguish different sub-categories that belong to the same super-category, such as birds [1], [2], dogs [3], cars [4], etc., has many important applications in a broad range of fields, including biodiversity monitoring [5], [6], transportation [7], security [8], [9], and retail [10]. However, it is still a challenging task due to the following issues. First, the visual similarities among the different sub-categories are high from the shape, color, texture, and other aspects, which makes it difficult to distinguish different sub-categories by the global features. Second, the images belonging to the same sub-category may have large variations caused by posture, angle, illumination, and background, which makes it difficult to extract consistent and robust feature representations.

To address the challenges in the FGVC task, the semantic parts and discriminative features are crucial for accurate classification [11]. A desirable and effective model should fully exploit the various discriminative part features and avoid the influence of noisy backgrounds to obtain robust and discriminative feature representation. Existing methods usually [12], [13], [14], [15], [16] utilize the class labels to locate the discriminative parts automatically without extra annotations. These methods can be roughly classified into three types: part localization-based, attention-based, and feature channel-based methods. Part localization-based methods need to utilize object detection techniques to locate the semantic parts in the image. For instance, Zhu et al. [15] proposed to extract texture features from the located and cropped rectangle regions in the image by the Gabor filters. Ma et al. [17] proposed to locate and crop the discriminative regions and re-input to the backbone for detailed information. However, the rectangular anchor boxes for localization do not adapt well to local regions with variable shapes, thus involving a lot of irrelevant background information.

The attention-based methods [18], [19], [20], [21], [22], [23], [24], [25] attempt to distinguish the objects or parts by learning the attention scores. For example, Zhang et al. [25] proposed to select several discriminative tokens through the transformer framework, where the core of vision transformer [19] is the multi-head self-attention, which captures the interrelationship between image patches by the attention weights. But each key and value feature in the attention mechanism only contains a single image patch, as shown in Figure 1(a). To solve this issue, Xia et al. proposed the deformable attention [23], as shown in Figure 1(b). Although

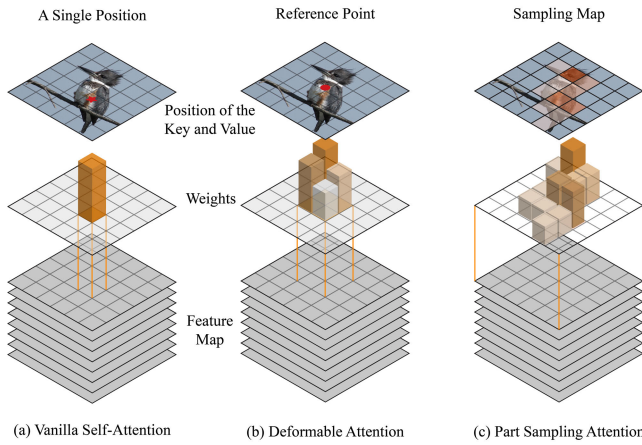


Fig. 1. Methods of obtaining key and value with different attention mechanisms. (a) the vanilla self-attention (b) the deformable attention (c) the proposed part sampling attention.

deformable attention captures the features surrounding each reference point, it still needs one layer to model the relationships of related parts over long distances. In brief, it is difficult to capture the dependencies of discriminative parts with various patterns via existing attention-based methods.

Feature channel-based methods [26], [27], [28], [29], [30], [31], [32], [33] proposed to utilize the semantic part information contained in the feature channels. Zhao et al. [29] proposed to learn the relationship between position and semantic features through feature channels and cluster higher-order relationships through semantic grouping. Wang et al. [33] proposed to construct structure-aware nodes through the channels of feature maps and explore implicit structure knowledge of the object with the structure-driven relation graph. However, these methods do not fully utilize the features of different levels and lack the combination of semantic features and part features.

To address the aforementioned issues, in this paper, we propose a novel multi-granularity part sampling attention (MPSA) network for fine-grained visual classification problem. The proposed MPSA network mainly consists of multi-granularity part retrospect (MPR), part sampling attention (PSA), and multi-granularity fusion (MF). To be specific, the proposed MPR block extracts the multi-granularity part features and supplements the high-level features with discriminative part clues. Based on the observation that the part features can be sampled from the feature map and weighted on the spatial dimension, we then propose the part sampling attention to learn the features from implicit semantic parts. The proposed PSA generates several sampling maps for the semantic parts and then performs detailed part sampling in the feature map to obtain the part features, as shown in Figure 1(c). Also, to ensure the diversity and robustness of the sampled parts, we adopt a part dropout strategy and utilize the importance of the sampled semantic parts to obtain the feature weights. We fuse the high-level features with the sampled part features to incorporate the part information clues. To further enhance the discriminative regions and suppress the noise from the background, we propose the multi-granularity fusion for fusing the features of multiple levels by the feature weights

to obtain more valuable feature representation. The gradient class activation map is used to supervise the feature weights to get more accurate feature weights. The proposed MPSA fully utilizes and fuses implicit semantic part information of different granularities and can be applied to various backbone networks. Experiments on four widely used fine-grained visual classification datasets demonstrate the effectiveness of the proposed MPSA method. The main contributions can be summarized as follows:

- We propose a novel MPSA network for fine-grained visual classification problem. MPSA fully explores the part features of different granularities for detailed and comprehensive feature representation.
- The proposed part sampling attention module samples and extracts the semantic part features with diverse shapes from the feature map.
- We propose the multi-granularity part retrospect block supplements the high-level feature map with multi-granularity semantic part information and clues.
- We propose the MF to aggregate the features of discriminative regions by the feature weights generated by the part features of all granularities.

The rest of the paper is organized as follows. Section II briefly reviews the related works. Section III describes the proposed method. Section IV reports the results of experiments and the comprehensive analysis. Finally, Section V presents the conclusions and the suggestions for further work.

II. RELATED WORK

In this section, we will briefly review the fine-grained image classification methods based on discriminative part learning from three categories: part localization-based, attention-based, and feature channel-based methods.

A. Part Localization-Based Methods

Part localization-based methods aim to explicitly locate discriminative parts and extract features via anchor boxes. Early works used manually annotated anchor boxes, such as [34], which extracted semantic parts by the annotations and input them into the backbone network to extract local features. However, expensive annotations limit the further development of these methods. Recent works mainly rely on weakly supervised learning to achieve part localization. For example, Yang et al. [35] utilized the FPN [36] network to locate discriminative parts and regularize the features through self-supervised pose alignment. Yang et al. [37] proposed to make coarse prediction by the global feature, and then refine the classification results with the located part features. Zhao et al. [24] explored the inherent higher-order correlation within each semantic part. Zhang et al. [25] proposed a two-stage network to extract global features and the cropped local features, respectively. However, it is difficult to adapt to discriminative parts with variable shapes by using rectangular anchor boxes for localization. The proposed part sampling method can adaptively adjust the sampling weights for each location based on the position of the implicit semantic parts in the feature map.

B. Attention Based-Methods

Methods based on attention mechanism typically use attention scores to locate discriminative features. For instance, Rao et al. [38] proposed the use of counterfactual attention to weaken the effect of background information on attention maps to obtain more accurate attention localization. Recently, many works based on Vision Transformer (ViT) have been applied to fine-grained image classification. The core of ViT is the self-attention mechanism that builds the interconnection between image patches. For example, He et al. [22] proposed to select a few of the most discriminative image patches for classification based on attention maps. Xu et al. [20] combined ensemble learning strategies with multi-head attention mechanisms to locate the most discriminative image patches. Hu et al. [39] used binary attention maps and the flood-fill algorithm to locate the discriminative region in the image. Liu et al. [40] proposed to learn invariant cues of the features through the attention maps of all stages in the network. However, attention mechanisms often ignore the features of other potential regions and only focus on a few regions and their surroundings. The proposed multi-granularity part retrospect and part sampling attention generate sampling maps at multiple levels to cover the semantic parts, thus obtaining the comprehensive and accurate part features.

C. Feature Channel-Based Methods

Feature channel-based methods generally use the channels of feature maps to obtain semantic parts information. Du et al. [31] proposed to constrain the consistency of the corresponding channel features of the feature maps of the same category in order to encourage the network to focus on the regions related to that category. Zheng et al. [28] proposed trilinear attention to model the relationships between channels and highlight the discriminative parts with high resolutions. Luo et al. [27] proposed to generate semantic parts with a regularization module that guides the feature maps to represent parts across different classes. Zhao et al. [29] proposed to learn the relationship between position and semantic features through feature channels and cluster higher-order relationships through semantic grouping. The pose-guided transformer proposed by Ma et al. [30] aggregates the information of the corresponding regions of the feature map using the heatmaps generated by a sub-network. Chen et al. [32] proposed to generate a decision mask by gathering the feature map and building relationships by the features of discriminative regions with GCN. However, these methods do not take full advantage of multi-level features, and the high-level feature maps lack detailed part features. To solve this problem, we propose multi-granularity part retrospect block, which supplements multi-granularity part information for the high-level feature map.

III. METHOD

This section introduces the overall framework of the proposed MPSA network in details. As shown in Figure 2. Given an input image, it is fed into the backbone network to get the feature maps. The extracted feature maps from all levels

are input into the MPR block. Each PSA module in the MPR block takes the feature maps of each level and the highest level feature map as input. The PSA modules comprehensively sample and learn the discriminative part clues to obtain the high-level feature maps enhanced with part information and get the feature weights. To highlight the foreground features and suppress the backgrounds, the feature maps and the feature weights are fed into the proposed MF to extract features for classification. The network is optimized by cross-entropy loss and the similarity loss with the supervision of the gradient map.

A. Part Sampling Attention

Most of the existing methods are based on the discriminative part features and usually extract the parts by anchor box localization and attention mechanism. The anchor box localization introduces irrelevant background information, and existing attention mechanisms only sample the individual position according to the weights of the attention map. So, they can not sample and extract the parts that vary in shape and ignore the parts of different importance. To this end, we propose the Part Sampling Attention (PSA) to sample implicit semantic parts of various shapes and the part features with long and short dependencies comprehensively. The process of PSA is illustrated in Figure 2.

1) *Part Sampling*: Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, we first feed it into the backbone network $f(I; \theta)$ and extract the feature map $F_i, i \in \{1, 2, \dots, s\}$ of each stage, where H, W represent the height, width of the image, where θ is the parameter of the network, and s denotes the total stages of the network. To sample various part features from a certain level, the input feature map $F_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ of i -th stage is fed into a linear layer followed by a softmax operation to obtain sampling map S_i which can be computed as,

$$S_i = \text{Softmax}(\text{FC}(F_i)) \in \mathbb{R}^{H_i \times W_i \times \frac{C_i}{r}} \quad (1)$$

where r represents the part sampling rate, H_i, W_i, C_i represent the length, width, and number of channels of the feature map, and $\text{FC}(\cdot)$ is the fully connected layer. Each dimension of the sampling map S_i represents the position of a certain semantic part and the corresponding weight, as shown in Figure 3. Then, the part features P'_i are sampled from the sampling maps S_i and feature maps as follows,

$$P'_i = \sum_x \sum_y^{H_i, W_i} (S_i + B_i)_{(x,y)} \odot F_{i(x,y)} \in \mathbb{R}^{\frac{C_i}{r} \times C_i} \quad (2)$$

where B_i represents the learnable bias parameter matrix, which can adapt to the region of the sampled parts at each stage. The learnable bias B_i is designed to guide the sampling map S_i in locating semantic parts. To be specific, during the training process, the learnable bias will obtain larger values at locations where the object appears with higher frequency. So, locations with higher values in the learnable bias usually represent the task-specific regions of the objects. It is worth noting that B_i contains task-specific regional information as part of the network's learnable parameters and will be reinitialized when the network is applied to other tasks.

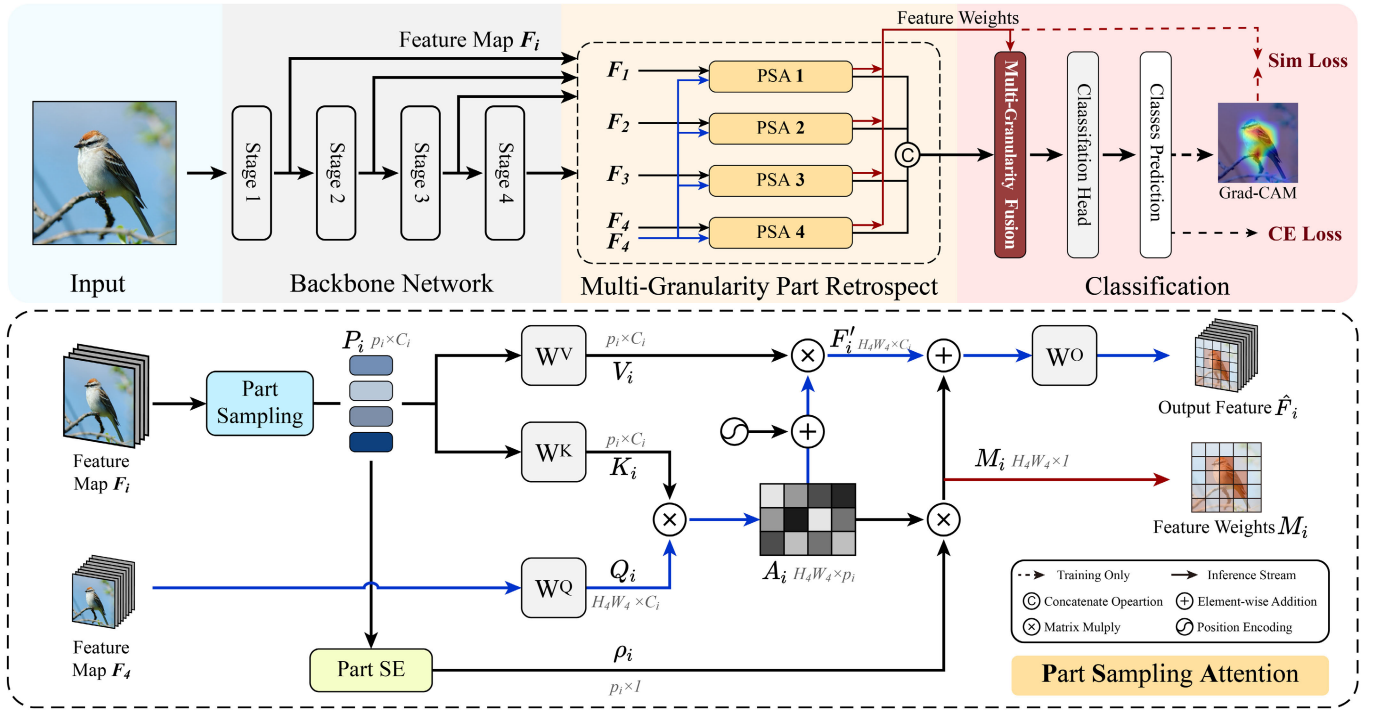


Fig. 2. The overview of the proposed Multi-Granularity Part Sampling Attention network. The upper part of this figure is the overall process of the network, and the lower part is a detailed explanation of part sampling attention.

To increase the robustness of the sampled part features, existing methods like [41] randomly drop the values in the feature, which fails to build a semantic correlation of the part features. Therefore, we apply a novel part dropout strategy by randomly discarding a portion of the sampled parts with the part dropout rate d to increase the robustness as well as the diversity of the sampled part features. The total number of part features p_i is reduced to $\lceil (1 - d)C_i/r \rceil$. The part features $P_i \in \mathbb{R}^{p_i \times C_i}$ can be obtained as,

$$P_i = \text{PartDrop}(P'_i) \in \mathbb{R}^{p_i \times C_i} \quad (3)$$

Compared to bilinear pooling [42], there are two key distinctions with Eq. 2. Firstly, the sampling map is obtained by linearly projecting and applying a softmax mapping on the feature map, thereby possessing non-linearity for a more precise characterization of various semantic parts. Secondly, the sampling map describes the shapes of semantic parts through weights assigned to individual positions, and samples from the feature map. This differs significantly from bilinear pooling [42], which establishes relations between channels.

2) *Part Attention*: The part attention is designed based on the cross-attention mechanism to supplement rich part features of each granularity for the high-level feature representation. Specifically, the high-level feature map $F_s \in \mathbb{R}^{H_s W_s \times C_s}$ and the sampled part features P_i are first projected by learnable parameters W_i^q , W_i^k , and W_i^v , to obtain $Q_i \in \mathbb{R}^{H_s W_s \times C_i}$, $K_i \in \mathbb{R}^{p_i \times C_i}$, and $V_i \in \mathbb{R}^{p_i \times C_i}$ as follows:

$$Q_i = F_s W_i^q, \quad K_i = P_i W_i^k, \quad V_i = P_i W_i^v \quad (4)$$

where s is the stage number of network. Then, the part attention and enhanced high-level feature map F'_i are calculated

through the following equations,

$$A_i = \frac{Q_i K_i^\top}{\sqrt{C_i}} + B'_i \quad (5)$$

$$F'_i = \text{Softmax}(A_i) V_i \quad (6)$$

where $B'_i \in \mathbb{R}^{H_s W_s \times p_i}$ represents the position encoding. Here, the multi-head attention mechanism is adopted when calculating Eq. 5 and Eq. 6. For brevity, we omit the relevant notations. By the proposed part attention, we can extract and supplement detailed part features from each stage and enhance the feature representation ability of the high-level feature representation.

3) *Part Squeeze-Excitation*: In order to highlight the salient objects, it is necessary to get the weights of feature maps at each spatial position. In the attention map, each column represents the attention scores of each part at different spatial locations. An intuitive approach is to treat the sum aggregation of attention weights for all sampled parts as feature weights. However, this does not consider the importance of sampled parts. The parts that represent the background information are not important for calculating attention scores. To address this issue, we propose a part squeeze-excitation module to learn the importance information for each part feature and combine it with the attention map to get the feature weights of each location in the feature map. Specifically, to get the importance $\rho_i \in \mathbb{R}^{p_i}$ for each part, the channel dimension of the part features are compressed and projected with two linear layers followed by sigmoid function, as shown in Figure 4. Different from the SE-Net [43] that squeezes the spatial dimension for the channel importance and re-weights the feature map, the proposed part squeeze-excitation module

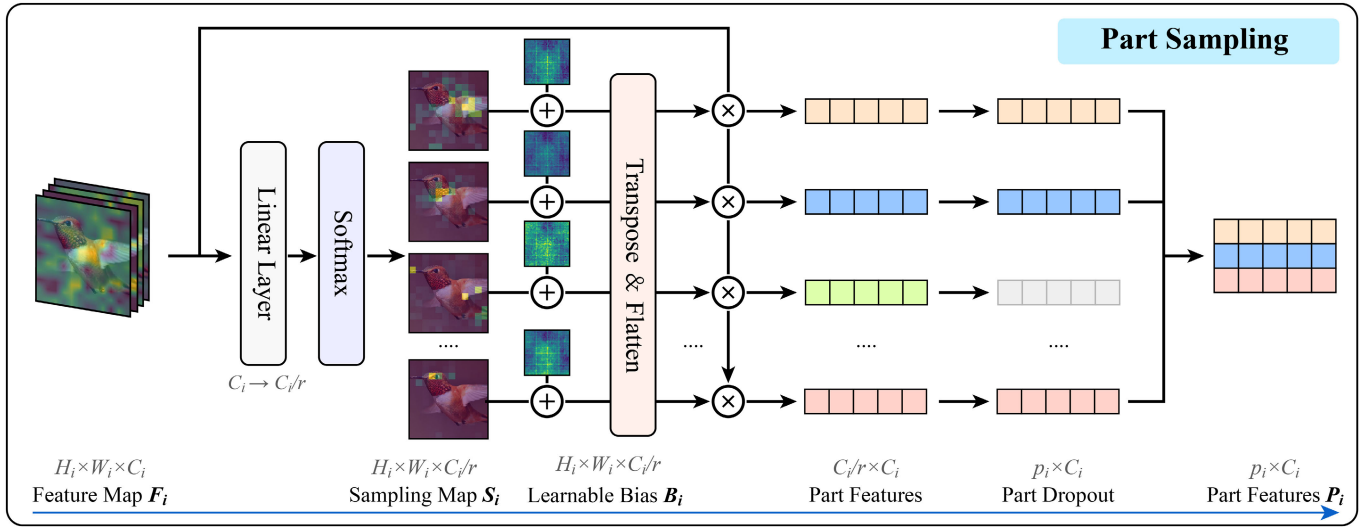


Fig. 3. The flow diagram of the part sampling. Firstly, a non-linear projection is used. The obtained part sampling map is added to the learnable bias. Then, multiplied with the feature map and a sum operation, the part features are obtained. To ensure feature robustness, the part dropout is applied at the stage.

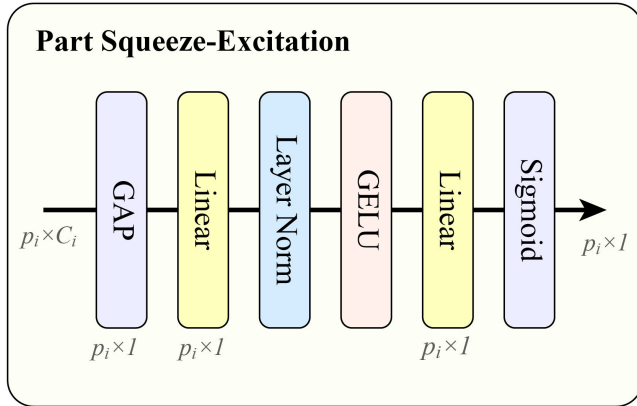


Fig. 4. The flow diagram of the part squeeze-excitation. The input part features are compressed through the channel dimension. Then, the compressed features are sequentially connected to the linear layer, the layers normalization, activation function, linear layer, and sigmoid function to obtain the importance scores for each part.

conducts a squeezing operation in the channel dimension for the part importance ρ_i .

Then, the attention map $A_i \in \mathbb{R}^{H_s \times W_s \times p_i}$ calculated by Eq. 5 is multiplied with the importance of each part ρ_i , and summed up in each column. The obtained feature weights M_i is represented as:

$$M_i = \sum_j^{p_i} (\text{Softmax}^*(A_i) \odot \rho_i)_j \quad (7)$$

where $\text{Softmax}^*(\cdot)$ represents the softmax function, which is applied to all rows to ensure that the summation of attention of each part across spatial locations is equal to 1. The learned feature weights are utilized to highlight the foreground objects. The output feature map $\hat{F}_i$ is obtained by the following equation,

$$\hat{F}_i = (F'_i + \alpha_i M_i) W_i^o \quad (8)$$

where W_i^o is the parameter of linear projection, and α_i represents the learnable scaling coefficient. Through part squeeze-excitation, the feature weights of each stage and features map supplemented with part clues are outputted.

B. Multi-Granularity Part Retrospect

The proposed PSA focuses on extracting the single granularity part features. To obtain the discriminative part features of multi-granularity and enhance the high-level feature representation with detailed part information, we propose the multi-granularity part retrospect block, which is composed of multiple PSA modules. The MPR block aggregates the output features of multiple PSA modules to get the feature maps that are supplemented with the part information of different levels and obtain the corresponding feature importance weights. The output multi-granularity feature map contains detailed part features of different granularities for the following feature fusion.

Specifically, the feature maps of all stages are fed into the MPR block, which includes s PSA modules. As shown in Figure 2, the i th PSA module takes the feature map of the i -th stage $F_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ and the high-level feature map $F_s \in \mathbb{R}^{H_s \times W_s \times C_s}$ as the inputs and outputs the enhanced feature map with part information $\hat{F}_i \in \mathbb{R}^{H_s \times W_s \times C_i}$ and the feature weights $M_i \in \mathbb{R}^{H_s \times W_s}$ of i th stage,

$$\{\hat{F}_i, M_i\} = \text{PSA}(F_s, F_i). \quad (9)$$

The high-level feature map and part features of different granularities are concatenated to obtain the multi-granularity feature map F^* as follows,

$$F^* = \sigma(\text{Concat}(\hat{F}_1, \hat{F}_2, \dots, \hat{F}_s)) \quad (10)$$

where σ represents the activation function. Through the proposed MPR block, the high-level feature map supplemented by part clues and the feature weights of each stage can be obtained.

Algorithm 1 Multi-Granularity Part Sampling Attention

Input: Input Image X , Backbone network f , Stages s , Hyperparameters r, d, α, β . Learnable Bias B_i , Projection weight W^o , Activation Function $\sigma(\cdot)$.

Output: Prediction logits z .

```

1:  $\{F_1, \dots, F_s\} = f(X)$ 
2: # Multi-Granularity Part Retrospect
3: for  $i$  in  $1, \dots, s$  do
4:   # Part Sampling Attention
5:    $P_i = \text{PartSampling}(F_i, B_i, r, d)$   $\triangleright$  Eq. 1,2,3
6:    $\rho_i = \text{PartSE}(P_i) \in \mathbb{R}^{p_i}$   $\triangleright$  Figure. 4
7:    $F'_i, A_i = \text{PartAttention}(F_s, P_i)$   $\triangleright$  Eq. 4,5,6
8:    $M_i = \sum_j^{p_i} (\text{Softmax}^*(A_i) \odot \rho_i)_j$   $\triangleright$  Eq. 7
9:    $\hat{F}_i = (F'_i + \alpha_i M_i) W_i^o$   $\triangleright$  Eq. 8
10: end for
11:  $F^* = \sigma(\text{Concat}(\hat{F}_1, \hat{F}_2, \dots, \hat{F}_s))$   $\triangleright$  Eq. 10
12: # Multi-Granularity Fusion
13:  $M^* = \sum_{i=1}^s \beta_i M_i$   $\triangleright$  Eq. 11
14: Return:  $z = \frac{1}{H_s W_s} \sum_x \sum_y^{W_s} (F^* \odot M^*)_{(x,y)}$   $\triangleright$  Eq. 12
```

C. Multi-Granularity Fusion

Most existing methods use a global average pooling for feature aggregation before image classification. However, these method neglect the importance of discriminative regions in feature fusion. Therefore, multi-granularity fusion (MF) is proposed, which can emphasize the importance of discriminative regions in feature fusion.

Because the feature weights M_i of different stages usually focus on different regions, MF uses all of them for weighting the multi-granularity feature map F^* output by the MPR block. In the MF, all the feature weights M_i of different levels are multiplied to the corresponding learnable parameter β_i to get the comprehensive feature weights M^* by the following equation,

$$M^* = \sum_{i=1}^s \beta_i M_i \quad (11)$$

The output feature z is calculated by the multi-granularity feature map F^* and multi-granularity fusion operation as follows

$$z = \frac{1}{H_s W_s} \sum_x \sum_y^{W_s} (F^* \odot M^*)_{(x,y)} \quad (12)$$

The overall progress of our proposed method is referred to Algorithm 1.

D. Loss Function

Finally, the output feature z is fed into a fully connected layer for classification, and the cross-entropy loss is adopted for optimization,

$$\mathcal{L}_{ce} = -\bar{y} \log(\text{FC}(\text{LN}(z))) \quad (13)$$

where, $\bar{y}$ represents the ground truth label, $\text{LN}(\cdot)$ is layer normalization [44].

Furthermore, as the utilization of multi-granularity fusion, the output feature z depends more on the feature weights. To obtain more accurate and robust feature weights, the gradient-weighted class activation map (Grad-CAM) [45], [46] is employed for supervision. Notably, the feature weights we optimized are used for the prediction. Specifically, the gradients of the multi-granularity feature maps with respect to the corresponding class logit y^c is $\frac{\partial y^c}{\partial F^*}$. Then, the Grad-CAM is computed as:

$$M_{CAM} = \sigma \left(\sum_{k=1}^{C'} \alpha_k F_k^* \right) \quad (14)$$

$$\alpha_k = \frac{1}{H_s W_s} \sum_i \sum_j \frac{\partial y^c}{\partial F_{ij}^*} \quad (15)$$

where $C' = \sum_{i=1}^s C_i$. We impose ℓ_2 distance constraint on the feature weights and obtain the similarity loss $\mathcal{L}_{cam}$ as follows:

$$\mathcal{L}_{cam} = \|M_{CAM}, M^*\|_2 \quad (16)$$

The overall loss of the network is combine the cross-entropy loss and similarity loss with a hyperparameter γ as follows:

$$\mathcal{L} = \mathcal{L}_{ce} + \gamma \mathcal{L}_{cam} \quad (17)$$

IV. EXPERIMENTS

In this section, we evaluate the proposed multi-granularity sampling attention (MPSA) on four commonly used FGVC datasets to validate the effectiveness of our approach. Firstly, we briefly introduce the datasets and the experimental settings. Next, we compare the results of MPSA with the state-of-the-art methods and perform the visual analysis of the experimental results. Finally, we conduct ablation experiments to verify the effectiveness of our method.

A. Datasets and Implementation Details

1) *Datasets*: The experiments are conducted on four popular FGVC datasets, i.e., CUB-200-2011 [1], Stanford Cars [4], Stanford Dogs [3], and NABirds [2]. As shown in Table I, we followed the official information to split training and testing samples. CUB-200-2011 is the most widely used bird dataset in FGVC, characterized by highly similar appearances among different bird species and significant intra-class variations due to variations in illuminations, angles, and poses. Stanford Cars consists of various types of vehicles, and its main challenge lies in the substantial visual differences caused by different car body colors for the same car model. Stanford Dogs includes 120 breed categories of dogs, and its main challenge is the presence of only partial views of the dog's body in the samples. NABirds is a large bird dataset that contains more classes and samples compared with CUB-200-2011.

2) *Implementation Details*: In all experiments, unless otherwise stated, we use the Swin Transformer Base model pretrained on ImageNet [47] as the backbone network for the extracted image features, same as [48]. The input images are first scaled to 512×512 pixels. During the training process, random cropping is applied, while during the testing

TABLE I

THE DETAILS OF DATASETS WE USED AND THE CORRESPONDING LEARNING RATES

Dataset	Class	Train	Test	LR
CUB-200-2011 [1]	200	5994	5794	5e-4
Stanford Dogs [3]	120	12000	8580	5e-5
Stanford Cars [4]	196	8144	8041	1e-4
NABirds [2]	555	23929	24633	5e-4

process, center cropping is used to obtain fixed-size images of 384×384 pixels. For the training process, random flipping and random Gaussian blur are applied to the CUB, Dogs, and NABirds datasets, and the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 is used as the optimizer. While random flipping, AutoAugment [49], CutMix [50], and Mixup [51] are applied to the Stanford Cars, and the AdamW [52] optimizer is used. The weight decay is set to $1e-8$ for all datasets. The initial learning rates for different datasets are shown in Table I. Referring to the fine-tuning settings of Swin Transformer [53], a cosine annealing learning rate scheduler with a 5-epoch linear warm-up is employed. The proposed model is trained for 50 epochs in total for all experiments.

In the MPR block, the feature maps of the last 4 stages are utilized. In the PSA, the part sampling rate r is set to 2.0, and the part dropout rate d is set to 0.4. The learnable weights α_i are initialized to 0.1. In the MF, the initial learnable weights β_i is set to 0.25. The loss control parameter γ is set to 1.0. We use GELU [54] as the activation function $\sigma(\cdot)$. Our network is implemented by PyTorch and trained with two Nvidia 3090 GPUs.

B. Comparison With State-of-the-Art Methods

1) *Results on the CUB-200-2011 Dataset:* As shown in Table II, the proposed MPSA achieves an accuracy of 92.8%, surpassing all existing methods. Compared to TransFG [22], our method achieves a notable improvement of 1.1% in accuracy, while TransFG requires extensive additional computation. Compared with PART [24], which also employs the attention mechanism to part information, our method achieves a significant improvement of 2.7%. The P2P-Net [35] utilizes the feature maps of each stage for curriculum learning and employs a detection network to obtain multi-scale part bounding boxes. The proposed PSA uses flexible sampling maps that capture part features at different scales. PSA also builds the correlations between the feature map and the part features and achieves an accuracy improvement of 2.6%. IELT [20] utilizes different attention heads to obtain diverse part features and extracts features by concatenating important tokens from different layers. However, this method compromises the spatial relationships of the output features. Our proposed MPR module supplements the high-level feature with multi-granularity part information, and the MF leverages the feature weights to enhance foreground features and achieves 1.0% accuracy improvement.

TABLE II

COMPARISON RESULTS ON CUB-200-2011

Method	Publication	Backbone	Accuracy (%)
RA-CNN [55]	CVPR 2017	VGG-19	85.3
SFFF [56]	TCSVT 2022	ResNet-50	85.4
Cross-X [27]	CVPR 2019	ResNet-50	87.7
PA-CNN [14]	TIP 2019	VGG-19	87.8
AP-CNN [13]	TIP 2021	ResNet-50	88.4
DSTL [57]	CVPR 2018	Inception-v3	88.8
FDL [58]	AAAI 2020	DenseNet-161	89.1
GaRD [29]	CVPR 2021	ResNet-50	89.6
PMG [59]	ECCV 2020	ResNet-50	89.6
GDSMP-Net [60]	PR 2023	ResNet-50	89.9
API-Net [61]	AAAI 2020	DenseNet-161	90.0
PMG-V2 [31]	TPAMI 2022	ResNet-101	90.0
PART [24]	TIP 2021	ResNet101	90.1
P2P-Net [35]	CVPR 2022	ResNet-50	90.2
DATL [62]	ISVC 2020	Inception-v3	91.2
CP-CNN [63]	TIP 2022	ResNet-50	91.4
LGTF [15]	ICCV 2023	DenseNet-161	91.5
TransIFC [40]	TMM 2023	Swin-B	91.0
CAMF [48]	SPL 2021	Swin-B	91.2
RAMS-Trans [39]	AAAI 2021	ViT-B 16	91.3
ViT-Net [64]	ICML 2022	Swin-B	91.6
TransFG [22]	AAAI 2022	ViT-B 16	91.7
IELT [20]	TMM 2023	ViT-B 16	91.8
MP-FGVC [65]	AAAI 2024	ViT-B 16	91.8
ACC-ViT [25]	AAAI 2024	ViT-B 16	91.8
DACL [66]	CVPR 2022	ResNet50-ViT	92.0
MPSA	-	Swin-B	92.8

2) *Results on the Stanford Cars Dataset:* In this task, as shown in Table III, our MPSA achieves an accuracy of 95.4%, reaching the current state-of-the-art performance. Compared to the CP-CNN [63], which achieves the same accuracy by employing multiple backbone networks to extract global and local part features separately, our proposed MPSA exhibits better scalability and avoids the burden of additional parameters and computational complexity. P2P-Net also achieves a result of 95.4% on the Cars dataset. Although P2P-Net [35] shares the backbone network parameters between the global and local branches, it still requires cropping multiple part regions and re-extracting the features, leading to increasing training time. In contrast, MPSA utilizes multi-level feature maps to extract part features instead of cropping regions. As a result, the input image only needs to pass through the backbone network once for the prediction, simplifying the training process compared to P2P-Net.

3) *Results on the Stanford Dogs Dataset:* As shown in Table IV, our MPSA achieves an accuracy of 95.4%, reaching the current state-of-the-art. Due to the characteristics of Stanford Dogs, where samples typically show only a part of the dog's body, methods that rely on predefined uniform-sized part features, such as TransFG [22], often fail to achieve outstanding results. The sampling map in the PSA can automatically sample maps to detect multi-granularity parts and extract features accordingly. Therefore, our MPSA performs well on the dog dataset with significant variations in part sizes.

TABLE III
COMPARISON RESULTS ON STANFORD CARS

Method	Publication	Backbone	Accuracy (%)
DSTL [57]	CVPR 2018	Inception-v3	90.6
RA-CNN [55]	CVPR 2017	VGG-19	92.5
PA-CNN [14]	TIP 2019	VGG-19	93.3
FDL [58]	AAAI 2020	DenseNet-161	94.0
SFFF [56]	TCSVT 2022	ResNet-50	94.4
DATL [62]	ISVC 2020	Inception-v3	94.5
Cross-X [27]	CVPR 2019	ResNet-50	94.6
GaRD [29]	CVPR 2021	ResNet-50	95.1
PMG [59]	ECCV 2020	ResNet-50	95.1
API-Net [61]	AAAI 2020	DenseNet-161	95.3
GDSMP-Net [60]	PR 2023	ResNet-50	95.3
PART [24]	TIP 2021	ResNet101	95.3
PMG-V2 [31]	TPAMI 2022	ResNet-101	95.4
AP-CNN [13]	TIP 2021	ResNet-50	95.4
P2P-Net [35]	CVPR 2022	ResNet-50	95.4
CP-CNN [63]	TIP 2022	ResNet-50	95.4
TransFG [22]	AAAI 2022	ViT-B 16	94.8
ViT-Net [64]	ICML 2022	Swin-B	95.0
CAMF [48]	SPL 2021	Swin-B	95.3
DACL [66]	AAAI 2022	ResNet50-ViT	95.3
MPSA	-	Swin-B	95.4

TABLE IV
COMPARISON RESULTS ON STANFORD DOGS

Method	Publication	Backbone	Accuracy (%)
FDL [58]	AAAI 2020	DenseNet-161	84.9
DSTL [57]	CVPR 2018	Inception-v3	85.2
RA-CNN [55]	CVPR 2017	VGG-16	87.3
SFFF [56]	TCSVT 2022	ResNet-50	88.0
Cross-X [27]	CVPR 2019	ResNet-50	88.9
API-Net [61]	AAAI 2020	ResNet-101	90.3
PMG-V2 [31]	TPAMI 2022	ResNet-101	90.7
LGTF [15]	ICCV 2023	DenseNet-161	92.1
DATL [62]	ISVC 2020	Inception v3	92.2
MP-FGVC [65]	AAAI 2024	ViT-B 16	91.0
IELT [20]	TMM 2023	ViT-B 16	91.8
TransFG [22]	AAAI 2022	ViT-B 16	92.3
RAMS-Trans [39]	AAAI 2021	ViT-B 16	92.4
CAMF [48]	SPL 2021	Swin-B	92.8
ACC-ViT [25]	PR 2024	ViT-B 16	92.9
ViT-Net [64]	ICML 2022	DeiT-B	93.6
MPSA	-	Swin-B	95.4

ViT-Net [64] utilizes a binary tree structure to classify discriminative features in the feature maps but fails to construct the interactions between features at different spatial positions. The proposed multi-granularity fusion assigns importance to different positions of the multi-granularity feature map, resulting in a 1.8% higher accuracy.

4) *Results on the NABirds Dataset:* We also compared the classification accuracy of NABirds, as shown in Table V. Our method achieves the highest classification accuracy of 92.5%, which demonstrates the effectiveness of our method on the large-scale dataset. Compared with TransIFC [40], our method

TABLE V
COMPARISON RESULTS ON NABIRDS

Method	Publication	Backbone	Accuracy (%)
Cross-X [27]	ICCV 2019	ResNet-50	86.2
DSTL [57]	CVPR 2018	Inception-v3	87.9
GaRD [29]	CVPR 2021	ResNet-50	88.0
API-Net [61]	AAAI 2020	DenseNet161	88.1
PMG-V2 [31]	TPAMI 2022	ResNet-101	88.4
GDSMP-Net [60]	PR 2023	ResNet-50	89.0
LGTF [15]	ICCV 2023	DenseNet-161	90.4
TransFG [22]	AAAI 2022	ViT-B 16	90.8
IELT [20]	TMM 2023	ViT-B-16	90.8
TransIFC [40]	TMM 2023	Swin-B	90.9
MP-FGVC [65]	AAAI 2024	ViT-B 16	91.0
ACC-ViT [25]	PR 2024	ViT-B 16	91.4
MPSA	-	Swin-B	92.5

leads by 1.6%, which only retains and utilizes the features of the discriminative region, our method obtains richer and more comprehensive part features through part sampling. MP-FGVC learns the text prompts to guide the visual features for classification with a two stage training scheme, while our method uses MPR to supplement part information for feature maps to obtain better results and trains the whole model end to end.

C. Visual Analysis

In this section, we conducted the visual analysis of the proposed MPSA, including the visualization of the sampling maps, feature weights and the feature representation.

1) *Visualization of the Sampling Map:* The core of the part sampling attention is to generate diverse sampling maps based on the characteristics of the samples. To verify that our network can generate multi-granularity part features, we conducted a visual analysis. As shown in Figure 5, each row represents the sampling maps generated by different stages for various samples. We show two sampling maps for the first, second, and third stages and one for the fourth stage, which represents the high-level semantics and usually covers the main region of the object. It can be observed that sampling maps of different granularities can focus on the detailed feature information of different regions. This is because the sampling maps aim to sample comprehensive part features. Thus, in different levels of PSA, the sampled part features usually focus on different regions to obtain richer semantic part feature representations. Moreover, as the sampling maps are not constrained by bounding boxes, they can capture long-range correlations. For example, in the visualization of cars, we can see that the sampling map in the first stage mainly focuses on edge texture information such as windows and headlights. While the sampling maps in the second and third stages concentrate on parts of different sizes. In the sampling map of the third stage, we can observe the relationship between wheel arches, and it samples the front fender and rear fender as part features.

Furthermore, we visualize the learnable bias for each stage in the part sampling attention, as shown in Figure 6. It can

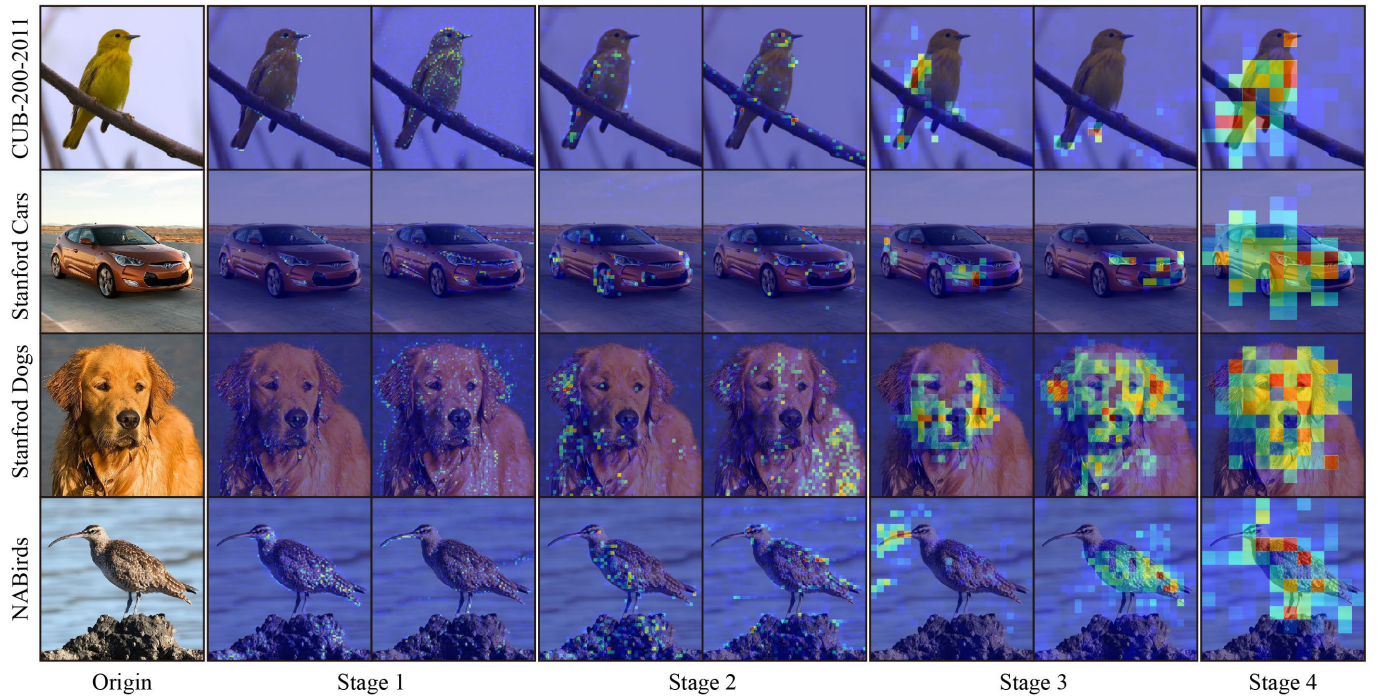


Fig. 5. Visualization of part sampling maps in the part sampling attention. The first column represents the original image, and each subsequent column represents the part sampling map at each stage. The red regions indicate higher values, while the blue regions indicate lower values.

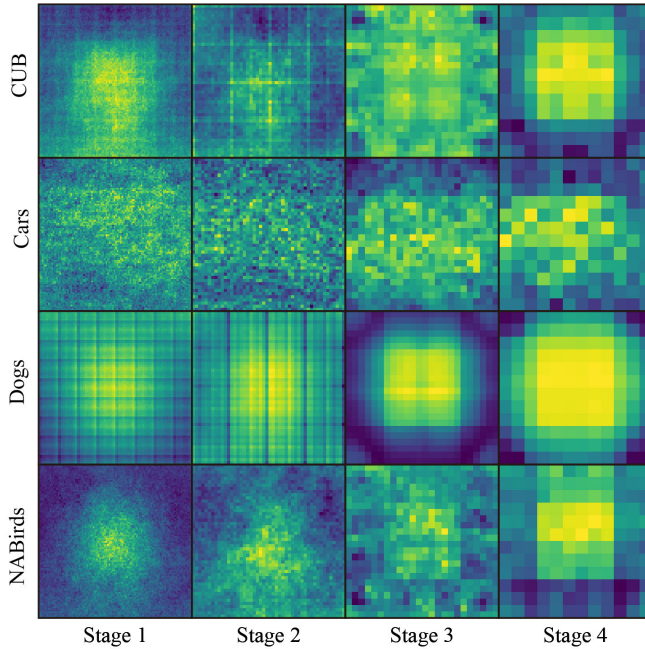


Fig. 6. Visualization of the learnable biases in part sampling attention.

be observed that the learnable biases of each dataset have their own characteristics. For example, bird samples tend to appear in the center of the image and stand upright, resulting in biases that favor the center position. On the other hand, vehicle samples typically appear horizontally in the center of the image, leading to biases that focus on the middle horizontal position. It is difficult to identify specific patterns in the biases of the dog dataset, and the biases for each stage tend to follow a Gaussian distribution, which we speculate is due to the

larger intra-class variations compared to the other datasets. Additionally, we notice that the biases of lower-stage feature maps are closer to the sample contours, while the biases of high-stage feature maps become more abstract, indicating that the network tends to let lower-stage sampling maps focus on contour and texture information, while higher-stage sampling maps concentrate on semantic information.

2) *Visualization of the Feature Weights*: To suppress the background area and highlight the foreground area, we propose multi-granularity fusion. As shown in Figure 7, we visualize the feature weights M^* and the gradient class activation maps with comparison with related works. It can be observed that the feature weights highlight the object region of the sample image and cover a wide range. Since the feature weights generated by PSA at each stage focus on different regions, the weighted feature weights are more likely to encompass the overall sample while highlighting the discriminative regions. By performing multi-granularity fusion with the feature maps, we obtain output features that contain rich information. For the Grad-CAM, which we used to supervise the feature weights, our method obtains an accurate response on the samples of each dataset. Compared with related works, the maps generated by our method obtain higher responses in the foreground regions of samples, especially in the discriminative regions. These results demonstrate that our proposed multi-granularity fusion can better highlight the foreground and suppress the background areas.

3) *Visualization of the Feature Representation*: The high-level features that are supplemented with the detailed part information of multi-granularities by MPR block can get better feature representation ability. In Figure 8, we use t-SNE to project the output features into the subspace and visualize the

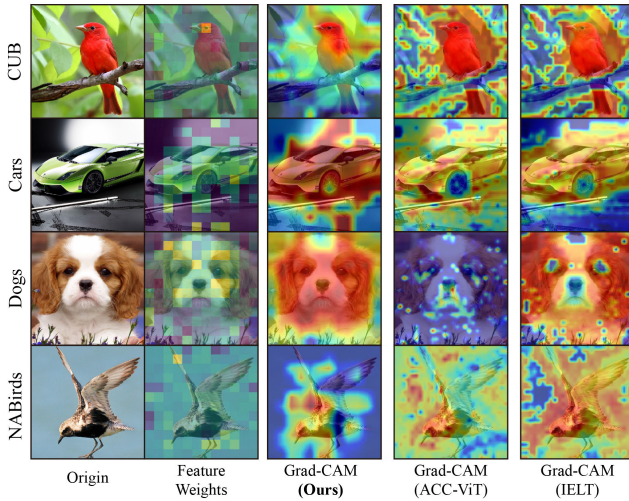


Fig. 7. Visualization of feature weights with Grad-CAM of our method, and comparison with related works. The first column represents the original image, the second column represents the feature weights, and the third to five column represents the Grad-CAM. In these visualizations, the red and brighter areas indicate higher values, while the blue and darker areas indicate lower values.

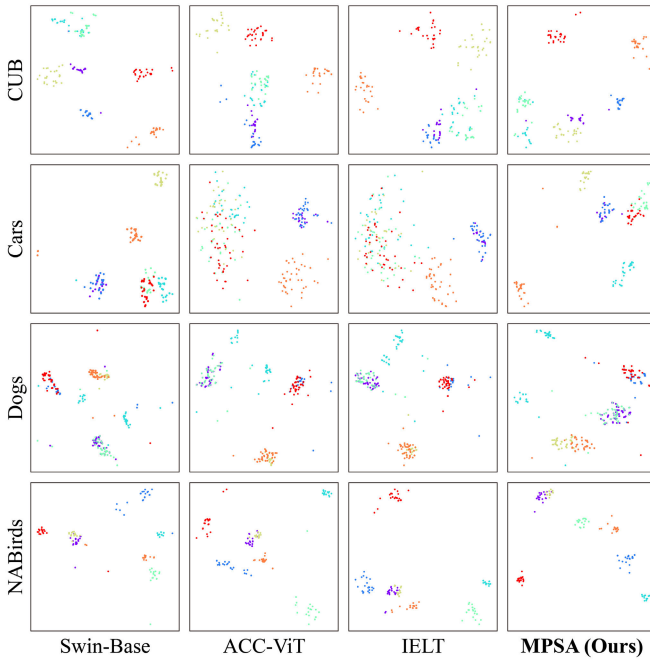


Fig. 8. The t-SNE visualization of all datasets and comparison with related works. Each dot represents a projected feature representation of a sample, and the color represents its category.

results of all datasets. Compared with the baseline method and related methods, it can be seen that our method has lower sample feature distances within each category, with higher inter-class margins in distinguishing different categories. This shows that the part features supplemented by PSA can improve the discriminative ability of features.

D. Ablation Studies

We conducted ablation experiments on MPSA to validate the effectiveness of the proposed module. Note that unless

TABLE VI
ABLATION EXPERIMENTS OF THE PROPOSED MODULES

Module			Accuracy (%)	Improv. (%)
PSA	MPR	MF		
-	-	-	91.93	-
✓	-	-	92.34	+0.41
✓	✓	-	92.67	+0.74
✓	-	✓	92.44	+0.61
✓	✓	✓	92.82	+0.89

otherwise stated, all ablation experiments are performed on the CUB-200-2011 dataset, as similar effects can be observed on other datasets as well.

1) *Component Analysis*: The ablation experiments are conducted on the three proposed modules, i.e., Multi-Granularity Part Retrospect (MPR) block, Part Sampling Attention (PSA), and Multi-Granularity Fusion (MF). The experimental results are shown in Table VI. Note that if the Multi-Granularity Part Retrospect block is not utilized, we only apply part sampling attention to the feature map of the highest level.

The PSA improves the accuracy by 0.41%, which can be attributed to its ability to capture diverse part features to fit part features of various shapes by the sampling maps. To supplement part information to higher-level feature maps, we introduce the MPR block. MPR block generates multi-granularity feature maps that contain parts of different levels of detail, leading to an accuracy improvement of 0.74%. Finally, by integrating the feature weights from different levels and performing multi-granularity fusion, the proposed MPSA reduces the impact of background noise and highlights the foreground region to achieve an improvement of 0.61% in accuracy.

Compared to the baseline without any modules, the method achieves an overall accuracy improvement of 0.89%, which relies on the synergistic cooperation of the three proposed modules. These experiments demonstrate the crucial contributions of the three modules in enhancing the classification results.

2) *Scalability Across Different Backbone Networks*: The additional experiments are conducted on different types of backbone networks, namely ResNet [67] and Vision Transformer (ViT) [19] to demonstrate the scalability of our proposed MPSA. Specifically, we use the pre-trained ResNet-50 and a ViT-B 16 as the backbone networks for experimentation. For both backbones, the input images are resized to 550×550 and then cropped to 448×448 while keeping the remaining experimental settings unchanged. Table VII shows the experimental results of MPSA combined with each backbone on the CUB, Stanford Cars, Dogs, and NABirds datasets. Similar to the results on the Swin Transformer backbone, our method achieves significant improvements when applied to ResNet and ViT. The enhancement is more pronounced in ResNet due to the introduction of long-range relationship modeling in convolutional networks. These experiments demonstrate that our proposed MPSA is applicable to hierarchical network designs such as Swin Transformer and ResNet, as well as fixed-scale network designs such as ViT. The consistent

TABLE VII

ABLATION STUDIES ON THE SCALABILITY OF DIFFERENT BACKBONES OF THE PROPOSE METHOD

Backbone	Method	Accuracy (%)			
		CUB	Cars	Dogs	NABirds
Swin-B	Baseline	91.93	94.67	93.76	92.09
	Ours	92.82	95.42	95.37	92.47
	Improv.	+0.89	+0.75	+1.61	+0.38
ResNet-50	Baseline	85.48	90.86	87.78	82.21
	Ours	87.47	93.25	90.19	86.52
	Improv.	+1.99	+2.39	+2.41	+4.31
ViT-B 16	Baseline	90.85	93.08	90.57	89.73
	Ours	91.23	93.74	91.46	89.97
	Improv.	+0.38	+0.66	+0.89	+0.24

TABLE VIII

INFLUENCE OF DIFFERENT PART SAMPLING RATE AND PART DROPOUT RATE

$r \backslash d$	0.0	0.2	0.4	0.6	0.8
1.0	92.46	92.48	92.42	92.39	92.44
2.0	92.39	92.60	92.82	92.61	92.46
3.0	92.63	92.42	92.54	92.44	99.37
4.0	92.49	92.42	92.54	92.51	92.58

accuracy improvements across the four datasets demonstrate the backbone-agnostic applicability and scalability of our method.

3) *Impact of Part Sampling Rate and Dropout Rate:* Our method introduces two hyperparameters in the part sampling attention, namely the part sampling rate r and the part dropout rate d . Table VIII presents the classification accuracy under different settings. It can be observed that as r and d increase, the number of generated sampling maps decreases, resulting in fewer part features in the part sampling attention and a reduction of the parameters in the network, and vice versa. We find that when both r and d keep low, the network does not achieve satisfactory performance. This is because without the dropout strategy, dominant parts with significant contributions tend to overshadow others, leading to a decrease in the diversity of sampling maps and part features. A large number of ineffective sampling maps introduce redundant parameters and irrelevant part features, which deviate from our design intent. By appropriately increasing the two rates, we can achieve a balance between accuracy and computational cost while enhancing the diversity of sampling maps and getting the accuracy improvement of 0.36%. However, when the part sampling rate and the part dropout rate are further increased, a majority of parts are discarded in each iteration, resulting in less stable classification results. Additionally, due to the higher part dropout rate, the repetition of sampling maps tends to increase to ensure the retention of some reliable part features after dropout. Finally, based on the experimental results, we select a part sampling rate of $r = 2$ and a dropout rate of $d = 0.4$.

TABLE IX

ABLATION STUDIES ON DIFFERENT USING OF PART IMPORTANCE FOR PART SAMPLING ATTENTION. FOR BREVITY, WE OMIT THE SUBSCRIPT i

How to Calculate $\hat{F}$ with ρ	Accuracy (%)
$\text{Softmax}(A)(\alpha\rho \odot V)$	92.65
$\text{Softmax}(A)(\alpha\rho + V)$	92.74
$\text{Softmax}(A)V \odot \alpha\text{Softmax}^*(A)\rho$	92.49
$\text{Softmax}(A)V + \alpha\text{Softmax}^*(A)\rho$	92.82

TABLE X

ABLATION STUDIES ON DIFFERENT FUSION METHODS

Method	Accuracy (%)
Global Average Pooling	92.67
Global Max Pooling	92.34
Spatial Attention [68]	92.21
Multi-Granularity Fusion w/o CAM	92.78
Multi-Granularity Fusion w/ CAM	92.82

4) Ablation Experiments With Part Importance Strategies:

The part sampling attention leverages the importance of parts that are obtained through part squeeze-extraction to generate feature weights, which are then fused with the feature maps to obtain the output. Table IX presents the experimental results using different strategies for incorporating part importance. It is worth noting that for the Hadamard product strategy, α is initially set to 1, while for the element-wise addition strategy, α is initially set to 0.1. Based on the experimental results, we observe that multiplying the normalized attention map by the part importance and then adding it to the feature maps can yield a maximum improvement of 0.33%. This could be attributed to the fact that calculating the feature weights by multiplying the part importance with the normalized attention map not only aligns better with our design intent but also ensures numerical stability. Furthermore, we find that the element-wise addition strategy outperforms the Hadamard product strategy. This may be because fluctuations in part importance have a smaller impact on the numerical values of the feature maps, thereby enhancing the stability of the network. Consequently, we select Strategy 4 as the method for incorporating part importance.

5) Ablation Experiments on Different Fusion Methods:

We propose multi-granularity fusion (MF) as an alternative to commonly used global average pooling (GAP) and global maximum pooling (GMP). Table X presents the experimental results for different feature fusion strategies. The spatial attention is proposed by CBAM [68], which concatenates the GAP and GMP results of the feature map and is followed by convolution and activation functions. Based on the results in Table X, our proposed MF achieves the highest performance and outperforms the commonly used GAP with a 0.15% increase in accuracy. The subpar performance of GMP may be attributed to the reduction in discriminative ability caused by the decrease in channel variance after applying maximum pooling to the multi-granularity feature maps. This may also affect the spatial attention strategy. Furthermore,

TABLE XI

ABLATIONS ON THE TRANSFER ABILITY OF THE LEARNABLE BIAS

Method	Accuracy (%)		
	CUB→Cars	CUB→Dogs	CUB→NABirds
w/ init.	59.59	88.85	88.52
wo/ init.	60.01	89.06	88.57

TABLE XII

THE COMPUTATIONAL COST ANALYSIS OF OUR METHOD WITH RECENT WORKS THAT BASED ON THE TRANSFORMER ARCHITECTURE. THE INPUT SIZE REPRESENTS THE HEIGHT AND WEIGHT OF THE INPUT IMAGE

Method	Backbone	Input Size	Param. (M)	FLOPs (G)	Memory (GB)
ViT [19]	ViT-B 16	448	86.4	78.5	1.54
RAMS-Trans [39]	ViT-B 16	448	86.4	157.4	2.53
TransFG [22]	ViT-B 16	448	86.4	130.2	1.44
IELT [20]	ViT-B 16	448	93.5	73.2	1.18
ACC-ViT [25]	ViT-B 16	448	87.0	162.9	2.01
Swin-Base [53]	Swin-B	384	87.1	47.2	1.19
ViT-Net [64]	Swin-B	448	92.2	65.6	1.40
MPSA (default)	Swin-B	384	97.4	49.2	1.28
MPSA	Swin-B	448	98.5	67.9	1.68

we conduct experiments on whether to apply Grad-CAM [45] for the supervision of feature weights. The result indicates that applying Grad-CAM leads to a slight improvement in accuracy. These experiments demonstrate the advantages of our proposed MF over other strategies.

6) *Transfer Ability of the Learnable Bias*: In order to verify the adaptability of the proposed learnable bias B_i in transfer learning, we verify the classification accuracy of the model trained on CUB when transferred to other datasets. We initialize the parameters of the classification head. Then, all parameters except B_i and the classification head are frozen, and the whole model is fine-tuned for 20 epochs. The experiment compares the impact of whether to initialize the learnable bias, and the results are shown in Table XI. Experimental results show that when the differences in the pre-trained models for downstream tasks are small, such as CUB→NABirds, whether to initialize the parameters has no significant impact on the classification results. On the contrary, when the downstream tasks are significantly different from the pre-trained task, such as CUB → Cars and Dogs, bias without initialization brings better results. This may be attributed to learnable biases adaptively learning part regions for downstream tasks with reference to prior knowledge, thereby achieving better results than random initialization. The above results demonstrate that our proposed learnable bias maintains excellent transfer ability even in tasks with huge differences.

E. Computation Cost Analysis

We further analyze the computational cost of our MPSA and then compare them with recent works [20], [22], [25], [39], [64] and the baseline models [19], [53], as shown in Table XII. Note that the default setting of the input image

TABLE XIII

THE COMPUTATIONAL COST ANALYSIS OF OUR METHOD WITH DIFFERENT SCALE OF BACKBONE NETWORKS

Backbone	Method	Acc. (%)	Param. (M)	FLOPs (G)	Memory (GB)
Swin-Tiny	Baseline	89.6	27.7	14.0	0.51
	Ours	90.1	34.1	15.0	0.58
Swin-Base	Baseline	91.9	87.1	47.2	1.19
	Ours	92.8	97.3	49.2	1.28
Swin-Large	Baseline	92.4	195.5	104.1	2.01
	Ours	92.9	216.2	109.3	2.20

size of our method is 384×384 . For fair comparisons, we also report the cost with the input size of 448×448 . From the results, we can observe that under default settings, our method only has 49.2G FLOPs, which is significantly better than other related methods. In terms of model capacity, the number of the parameters of our method is 97.4M, which is slightly higher than ViT-Net (92.2M). Under the same input size, the FLOPs and memory cost of our method are slightly higher than ViT-Net but still significantly lower than RAMS-Trans, TransFG, and ACC-ViT. Our method mines the part features comprehensively, so it is not necessary to use a large input size.

We also compare the differences in accuracy and computational cost between our method and baseline models of different scales in Table XIII. We have the following observations from the results: 1) Compared with the computational cost of the baseline, the additional cost introduced by our method only accounts for a small proportion of the total computational cost. Specifically, our method increases the FLOPs by 4% to 7%, the number of parameters by 11% to 23%, and the memory cost by 8% to 14% under different scale backbone settings. 2) Compared with baselines, the accuracy of our method is improved at all backbone scales. 3) The accuracy of our method on Swin-Base is improved by 0.4% compared with the Swin-Large baseline and has a lower number of parameters and memory cost. This further demonstrates the effectiveness of our method. Overall, our method maintains good performance in terms of computational cost.

V. CONCLUSION

In this paper, we propose a multi-granularity part sampling attention network for fine-grained visual classification. We fully utilize and fuse implicit semantic part information of different granularities to supplement high-level feature maps through the multi-granularity part retrospect block. The proposed part sampling attention generates diverse sampling maps from the feature maps and extracts part features of various shapes comprehensively, and captures their correlations. We also obtain the importances of parts features and integrate them with attention maps to obtain the feature weights. Then, to highlight foreground regions and suppress background noise, our proposed multi-granularity fusion emphasizes important regions in the feature map by the feature weights. Experimental results on the four prevailing

fine-grained visual classification benchmarks demonstrate the superiority of our approach. In future work, we will explore lightweight models and extend the application of our method to other practical applications.

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